

The Microscope and the Discovery of the Cell

The eye can only see things above a certain size. Lenses and the microscope opened up a hidden world.

How we learnt to see the invisible

The human eye can only see objects above a certain size. Long ago, people found that a curved piece of glass — thick in the middle, thin at the edges, shaped like a lentil seed — could make small things look bigger. They called it a **lens** (from the word for lentil).

Activity 2.1: a round-bottomed flask full of water, placed over a book, makes the letters look larger — the water acts like a magnifying glass.

Two key scientists

Scientist	Year	Contribution
Robert Hooke	1665	Book <i>Micrographia</i> ; saw boxes in a slice of cork and named them cells — first use of the word "cell".
Antonie van Leeuwenhoek	1660s	Made better lenses; first to see bacteria and blood cells. Called the Father of Microbiology .

Magnification — how much bigger?

- Hooke's microscope: about **200–300 times**.
- An ordinary light microscope: about **100–400 times**.
- An **electron microscope**: about **10,00,000 (one million) times** — shows tiny parts inside the cell.

Cell = basic unit of life. Hooke's word stuck: every living thing is made of one or more cells.

Where students lose marks

Trap 1 — Who saw what. Hooke *named* the cell (from cork, which were dead, empty boxes). Leeuwenhoek first saw *living* microbes — he is the Father of Microbiology.

Trap 2 — Cork cells were not alive. Hooke saw empty compartments in dead cork; he did not see living cells.

Trap 3 — Magnification mix-up. Light microscope $\approx 100\text{--}400\times$; electron microscope $\approx 10,00,000\times$. Don't swap them.

Model answers — write them like this

Example A. Why is Leeuwenhoek called the Father of Microbiology?

Step 1: He made improved, more powerful lenses/microscopes.

Step 2: He was first to clearly see and describe tiny living things such as bacteria.

∴ For first observing living microbes, he is called the Father of Microbiology.

Example B. Why is the word "cell" linked to a honeycomb?

Step 1: Hooke saw many small, empty, box-like spaces in cork.

Step 2: They reminded him of the compartments of a honeycomb, so he called each a "cell".

∴ The honeycomb-like compartments gave the cell its name.

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